



HIV-1 drug resistance and genotypes circulating among HIV-positive key populations in Bangladesh: 2016 update

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ABSTRACT

Objective: HIV-1 subtyping data of Bangladeshi strains are available in global HIV Sequence Database up to 2007, and there is no sequence of drug resistance profile based on the *pol* gene segment. This study aimed to update HIV genotyping data and describe the drug resistance mutations for the first time from Bangladesh using specimens from the latest HIV sero-surveillance conducted in 2016.

Study design and methods: During HIV sero-surveillance, a total of 1268 people who inject drugs (PWID) and 3765 female sex workers (FSW) were screened and among them, 230 (18.1%) PWID and 7 (0.2%) FSW were HIV positive. Among HIV positives, randomly selected 74 specimens (60 male-PWID, 7 female-PWID, and 7 FSW) were subjected to *gag*, *pol*, and *env* gene sequencing using gene-specific primers. Genotyping was decided based on the partial *gag* and *env* genes while transmission dynamics was based on the *gag* sequence ($n = 237$). Drug resistance profiles were obtained by using the algorithm of the established available drug resistance database.

Results: HIV subtype C and C-related recombinants have remained the major circulating genotypes in Bangladesh. Although the recurring transmission of subtype C occurred among PWID, we identified possible transmission to other key populations (KPs), which suggests spillover from PWID through the sexual route. The prevalence of drug-resistant mutation was low, and all strains were susceptible to NRTIs and NNRTIs drugs. Unique recombination forms (URF) with genotype C for *gag-pol* and A1 for *env* was also identified.

Conclusions: The study findings warrant continuous monitoring of HIV-positive individuals and future investigation to identify social networks within and between KPs to halt the transmission and prevent new infections.

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Introduction

Bangladesh is still considered a low HIV prevalence country, but it is one of the four countries in the Asia Pacific region where new infections continue to increase (NSP, 2016). A sentinel surveillance system was established in 1998 to monitor the trends of HIV infection and risk behaviors for transmission among key populations (KPs), which include sex workers (male and female), people who used drugs, males having sex with males (MSM), and transgenders. The 9th round of this surveillance was conducted in 2011, which reported 1.1% HIV prevalence in KPs (NASP, 2011). However, the southern part of the capital city Dhaka experienced a

concentrated HIV epidemic (Azim et al., 2008) with 7.3% among People Who Inject Drug (PWID). In the same population, 9.9% HIV prevalence was found in 2015 (Alam et al., 2016), which significantly increased to 27.3% ($p < 0.001$) in 2016. HIV prevalence among PWID in the northern part of Dhaka also increased significantly from 1.2% in 2011 to 8.9% in 2016 ($p < 0.001$) (NASP, 2017). The current estimated number of people living with HIV (PLHIV) in Bangladesh is 14,000, with 919 newly recorded cases in 2019 (AIDSdatahub, 2019). This increase was observed in the presence of a Needle and Syringe Program (NSP), which has been operating since 1998. Data from the HIV behavioral surveillance survey (HIV-BSS) conducted in 2016 among PWID in Dhaka city showed that 53% of PWID shared needles and syringes in the last week as compared to 61% in 2007 (NASP, 2017). This clearly indicates that needle-sharing practices have not changed among this population segment over the past decade despite the ongoing NSP. Along with needle/syringe, PWID also performs unprotected

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sexual intercourse with their nontransactional and transactional female sex partners where condom use ranges from 24% to 55% (NASP, 2017).

Sequence data are very important to track HIV transmission within and between populations (Gilbert et al., 2007; Haaland et al., 2009; Taylor et al., 2008). A previous study based on partial *gag* sequence (proviral) showed that HIV strains identified in Bangladesh exhibited an extraordinary degree of genetic diversity but subtype-C was identified as predominant and mainly circulates within PWID, which might have been imported from India (Sarker et al., 2008). Other circulating recombinant forms (CRF) and pure subtypes such as A1, B, D, and G were also detected among KPs, which were genetically related to strains from countries such as India, Myanmar, China, Africa, and Middle East (Bontell et al., 2013; Sarker et al., 2011; Sarker et al., 2008). As the surrounding HIV-1 epidemic in India and China differs significantly (Takebe et al.,

2010), understanding the current epidemic by analyzing circulating HIV genotypes in KPs, particularly PWID, is very important.

Bangladesh is providing HIV treatment, care, and support since 2005 in collaboration with the corporate sector and NGO, which was expanded under the Global Fund through Save the Children up to 2012. The government of Bangladesh took over the procurement of anti-retroviral drugs (ARV) through the central medical storage depot since 2012 under the Health Nutrition Population Sector Development Programme. National AIDS/STD program (NASP) also provided support to ensure treatment services through NGOs involvement up to 2017. Finally, NASP had taken over whole services of treatment care since October 2017 (ASP (AIDS/STD Programme), 2018) and provide treatment with a preferred regimen of Tenofovir (TDF) + Lamivudine (3TC) or Emtricitabine (FTC) + Efavirenz (EFV). HIV *pol* gene sequence is used to identify mutations associated with resistance to reverse transcriptase (RT)

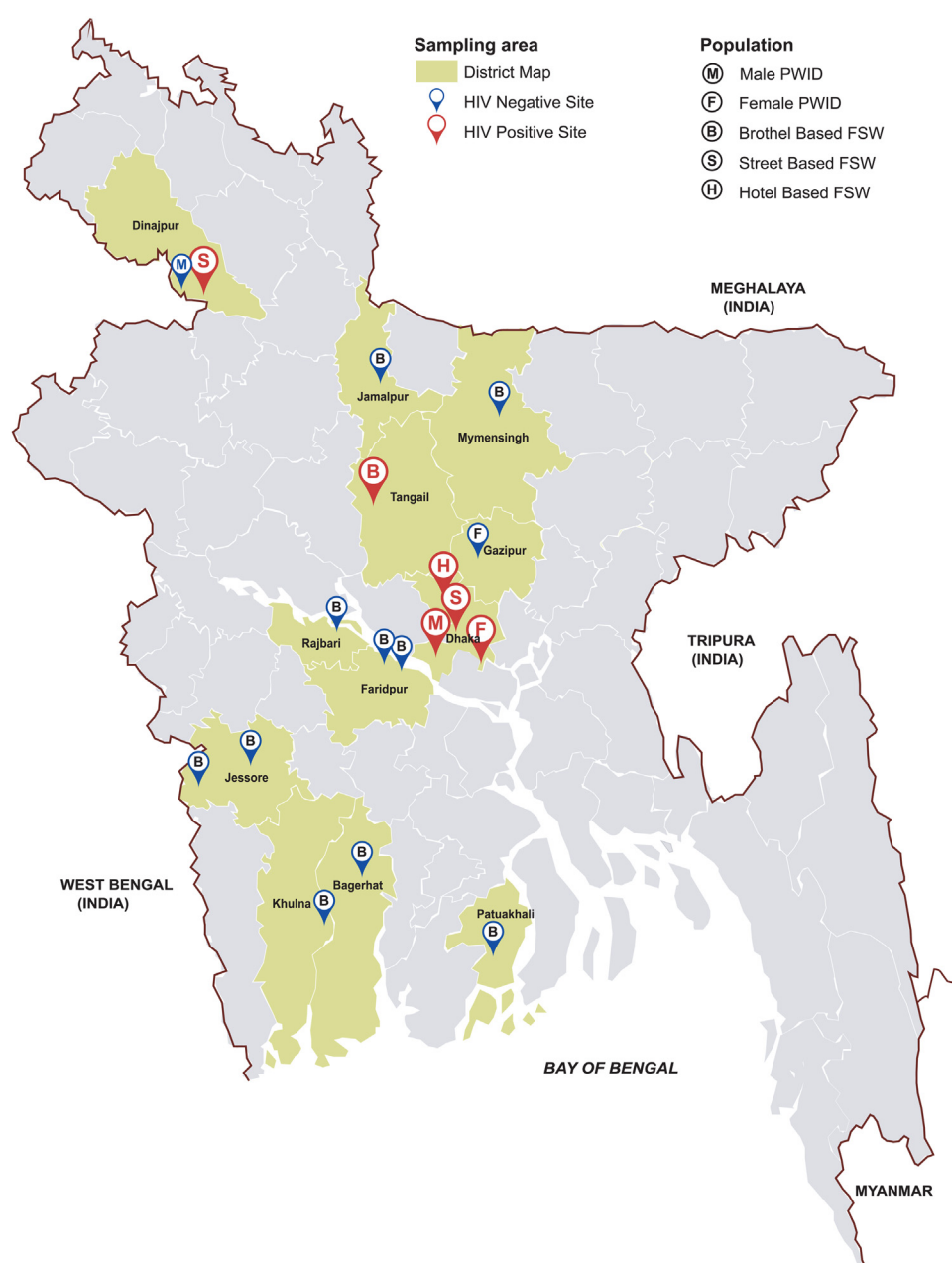


Figure 1. Study site from where samples were collected.

or the protease (PR) region (Pasquier et al., 2001). Most of the current knowledge on ARV resistance is based on subtype B of HIV-1, which only accounts for 10% of the global HIV infections (Santos and Soares, 2010) and 2% of the Bangladeshi strain (NASP, 2017). Viral suppression data are available in limited numbers while HIV drug resistance data are still not available, which is a major barrier to achieve the global 90–90–90 target, particularly the last “90.” Thus, the present study was designed to update HIV genotyping data and describe drug-resistant mutations available in Bangladesh.

Method

Study site, population, and sampling

The 10th National HIV Surveillance in 2016 was conducted among people who inject drugs (PWID) and female sex workers (FSW). This surveillance study was approved by the Govt. of Bangladesh and icddr,b Institutional Review Board (IRB). Geographical areas were limited to Hili (Dinajpur) and Dhaka for PWID and street-based FSW. Hotel- and resident-based FSW were included from Dhaka, while brothel-based FSW were included from all brothels of Bangladesh (Figure 1). A total of 5033 blood samples were obtained from several study sites, where 1268 PWID and 3765 FSW were enrolled. Most of the PWID were male from the capital city, Dhaka, which was divided into two regions: old Dhaka (southern part, n = 721) and rest of Dhaka (northern part, n = 291). In addition, 117 male PWID were enrolled from north-west land port area Hili, Dinajpur and 139 female PWID from Dhaka, Tongi, and Gazipur. A total of 1670 brothel-based sex workers (BBSW) were enrolled from 11 brothels located in nine districts in Bangladesh (Figure 1). Rest of the FSW were from Dhaka, 256 hotel-based sex workers (HBSW) and 1141 street-based sex workers (SBSW) and Hili, Dinajpur (197 SBSW).

Blood specimen (whole blood and serum) were obtained with informed and written consent from all participants. All specimens

were transported to the virology laboratory, icddr,b by maintaining a cold chain and initially screened for HIV antibody by a commercially available ELISA kit (Murex HIV Ag/Ab Combination, DiaSorin S.p.A. Dartford, UK). HIV antibody-positive results were later confirmed by a line immunoassay (LIA) (INNO-LIA HIV I/II Score, FUJIREBIO Europe N.V, Gent, Belgium). As previous data showed that Subtype-C is a major circulating genotype among male PWID, we presume that all risk factors and possible route of infection will remain same for male PWID. Therefore, among HIV-positive male PWID (n = 230), 60 specimens (26.1%) were selected randomly. As the number of positive cases for female PWID and FSW were low, we did not go for sampling, rather all specimens from female PWID (n = 7) and FSW (n = 7) were included for genotyping.

Gag and env gene amplification

Proviral DNA was extracted from whole blood samples using the DNeasy Blood & Tissue Kit according to the manufacturer's protocol (QIAGEN, Leusden, Netherlands). HotStarTaq Master Mix kit (QIAGEN, Leusden, Netherlands) with specific primers (Table 1) were used to amplify partial gag and env gene from proviral DNA by the nested PCR method.

Our primary target for the gag gene was to amplify 800 bp (nucleotide position 800–1600 based on HXB2 numbering), and we selected Primer 530F and CS4R (779–1822) for primary strand synthesis followed by nested PCR with 531F and CS2R (796–1631) or 532F and CS1R (836–1585). An alternative strategy was followed if nested PCR failed to generate any product. For this strategy, we used 0682F and 3885R (682–3885) primers for primary strand synthesis and 0776F and 3prot1R (776–2734) primers for nested PCR. With the secondary PCR product, we conducted 3rd round amplification with 530F and CS4R (779–1822) or 531F and CS2R (796–1631) primers.

For env gene, our primary target was to amplify 700 bp, nucleotide position 6800–7500. Primers ED5F and ED12R (6517–

Table 1
List of primers used in this study.

Primer		Position ^a	Name	Sequence (5'–3')	Reference
gag	Forward	0682–0708	0682F	TCTCTCGACGCAGGACTCGGCTTGCTG	Grossmann et al. (2015)
		0776–0800	0776F	CTAGAAGGAGAGAGAGATGGGTGCGAG	Grossmann et al. (2015)
		0779–0797	530F	GACTAGCGGAGGCTAGAAG	Sarker et al. (2008)
		0796–0818	531 F	GCGAGAGCGTCAGTATTAAG	Sarker et al. (2008)
		0836–0857	532F	GGGAAAAAATTCGGTTAAGGCC	Sarker et al. (2008)
	Reverse	1610–1585	CS1R	ACTATTTTATTTAATCCCAGGATTAT	Sarker et al. (2008)
		1658–1631	CS2R	TTTGGTCCTTGCTCTTATGTCCAGAATGC	Sarker et al. (2008)
		1844–1822	CS4R	CCCTGACATGCTGTCATCATTTTC	Sarker et al. (2008)
		5550–5574	5550F	AGARGAYAGATGGAACAAGCCCCAG	Grossmann et al. (2015)
		5861–5884	5861F	TGGAAGCATCCRGAAGTCAGCCT	Grossmann et al. (2015)
env	Forward	6517–6542	ED5F	ATGGGATCAAAGCCTAAAGCCATGTG	ANRS (2015)
		6559–6585	6559F	GGGATCAAAGCCTAAAGCCATGTGTA	Grossmann et al. (2015)
		6798–6816	ED31F	CAGTACAATGTACACATGG	ANRS (2015)
		7373–7350	7373R	GAAAAATTCTCTCYACAATTAAA	Grossmann et al. (2015)
		7504–7487	ED633R	GGAGGGGGCATACATTGC	ANRS (2015)
	Reverse	7771–7742	ED12R	AGTGCTTCTGCTGCTCCAAGAACCACAG	ANRS (2015)
		8445–8424	8445R	CTCTCTCCACCTTCTTCTTC	Grossmann et al. (2015)
	Forward	1810–1838	NonBF 1810s	GCTACACTAGAAGAAATGATGACAGCATG	Personal communication
		2002–2029	F2001s	TGCAGGCCCTAGGAAAAAGGGCTGTT	Personal communication
		2082–2109	5Prot1F	TAATTTTATAGGGAAGATCTGGCCTTCC	ANRS (2015)
pol	Reverse	2136–2163	5Prot2F	TCAGACGACACCAAGCCACAGCCCCA	ANRS (2015)
		2480–2499	MJ3	AGTAGGACCTACACCTGTCA	ANRS (2015)
		2530–2558	A35F	TTGGTTGACACTTAAATTTTCCATTAGTCC	ANRS (2015)
		2650–2621	3Prot2R	AATGCTTTTATTTTCTCTGTCAATGGC	ANRS (2015)
		2734–2703	3Prot1R	GCAAACTGAGTATTGTATGGATTTT	ANRS (2015)
	Reverse	3334–3300	NE1	CCTACTAAGCTTCTGTATGTCATTGACAGTCCAGCT	ANRS (2015)
		3420–3399	MJ4	CTGTAGTGTCTTGGTTCCTCT	ANRS (2015)
		3885–3867	3885R	CTGCTCCATCTACATAGAA	Grossmann et al. (2015)

^a Nucleotide position was based on HXB2 reference strain.

Table 2
Genotyping result for sequenced samples.

Risk group ^a	DIC ^b	ID ^c	Age ^d	Gag genotype ^e	Gag BLAST search ^f	Env genotype ^e	Env BLAST search ^f
SBSW	Hili, Dinajpur	HIV-05	32	C	India	CRF08_BC	China
HBSW	Kadom	HIV-74	40	CRF02_AG	Ghana	CRF11_CPX	Cameroon
SBSW	Khilgaon	HIV-73	20	C	Ethiopia	C	Malawi
SBSW	Pollobi	HIV-70	32	C	India	A1	India
BBSW	Kandarpara, Tangail	HIV-10	30	CRF62_BC	China	CRF08_BC	China
PWID_F	Bakshibazar	HIV-64	25	CRF62_BC	India		
PWID_F	Doyaganj	HIV-67	56	C	Bangladesh		
PWID_F	Doyaganj	HIV-68	31	C	India	CRF08_BC	China
PWID_F	Doyaganj	HIV-69	40	CRF07_BC	China		
PWID_F	Moulavibazar	HIV-63	29	CRF62_BC	Myanmar	CRF08_BC	India
PWID	Bakshibazar	HIV-33	35	CRF62_BC	China	C	India
PWID	Bakshibazar	HIV-40	42	CRF62_BC	China	CRF64_BC	Botswana
PWID	Bakshibazar	HIV-52	32	CRF62_BC	China	CRF08_BC	India
PWID	Bakshibazar	HIV-55	36	C	China	CRF64_BC	Nepal
PWID	Bakshibazar	HIV-56	32	C	India	CRF07_BC	India
PWID	Bakshibazar	HIV-58	38	C	Bangladesh	CRF08_BC	India
PWID	Bakshibazar	HIV-59	42	CRF62_BC	Bangladesh	CRF07_BC	China
PWID	Doyaganj	HIV-06	28	C	Bangladesh	CRF08_BC	China
PWID	Doyaganj	HIV-07	31	C	Bangladesh	CRF08_BC	China
PWID	Doyaganj	HIV-08	35	C	China	CRF08_BC	China
PWID	Doyaganj	HIV-09	37	C	India	CRF07_BC	Tanzania
PWID	Hajaribagh	HIV-41	30	C	India	CRF08_BC	China
PWID	Jatrabari	HIV-13	35	C	Ethiopia	CRF08_BC	India
PWID	Jatrabari	HIV-14	32	C	India	CRF08_BC	China
PWID	Jatrabari	HIV-16	45	C	India	CRF07_BC	India
PWID	Jatrabari	HIV-17	35	C	India	CRF08_BC	India
PWID	Jatrabari	HIV-18	45	C	China	CRF08_BC	India
PWID	Jatrabari	HIV-19	35	C	Ethiopia	CRF08_BC	China
PWID	Jurain	HIV-25	40	CRF62_BC	Bangladesh	CRF64_BC	Zambia
PWID	Kawranbazar	HIV-12	38	C	Ethiopia	CRF57_BC	Zambia
PWID	Khilgaon	HIV-01	30	C	Myanmar	CRF08_BC	China
PWID	Khilgaon	HIV-02	32	C	Bangladesh		
PWID	Khilgaon	HIV-03	30	C	Bangladesh	CRF08_BC	China
PWID	Khilgaon	–04	28	C	China	CRF08_BC	China
PWID	Khilgaon	HIV-11	35	C	China	CRF08_BC	China
PWID	Moulavibazar	HIV-35	35	C	China		
PWID	Moulavibazar	HIV-39	35	C	Myanmar		
PWID	Moulavibazar	HIV-42	55	C	China	CRF08_BC	Nepal
PWID	Moulavibazar	HIV-43	35	C	Bangladesh	C	China
PWID	Moulavibazar	HIV-47	38	C	Bangladesh	CRF64_BC	China
PWID	Nayabazar	HIV-36	45	C	Bangladesh	CRF08_BC	China
PWID	Nayabazar	HIV-37	46	C	India	C	China
PWID	Nayabazar	HIV-38	35	C	Bangladesh	CRF08_BC	China
PWID	Nayabazar	HIV-48	33	C	China	CRF08_BC	India
PWID	Nayabazar	HIV-49	38	C	China		
PWID	Nayabazar	HIV-51	35	C	Bangladesh	CRF08_BC	India
PWID	Nayabazar	HIV-53	35	C	Bangladesh	CRF08_BC	Nepal
PWID	Nayabazar	HIV-54	47	C	Myanmar	CRF61_BC	India
PWID	Nayabazar	HIV-57	35	C	Bangladesh	CRF08_BC	Zambia
PWID	S.K. Das	HIV-44	40	C	Ethiopia	CRF08_BC	China
PWID	S.K. Das	HIV-45	37	C	Bangladesh	CRF08_BC	China
PWID	S.K. Das	HIV-46	27	CRF62_BC	Bangladesh	CRF08_BC	Nepal
PWID	S.K. Das	HIV-50	38	C	Bangladesh	CRF08_BC	China
PWID	S.K. Das	HIV-60	53	C	India	CRF08_BC	China
PWID	S.K. Das	HIV-61	38	C	Bangladesh	CRF07_BC	China
PWID	S.K. Das	HIV-62	40	C	China	CRF08_BC	China
PWID	Sadarghat	HIV-20	32	C	Bangladesh	CRF64_BC	India
PWID	Sadarghat	HIV-21	43	C	Bangladesh	CRF08_BC	India
PWID	Sadarghat	HIV-23	35	C	Bangladesh	CRF08_BC	China
PWID	Sadarghat	HIV-24	42			CRF08_BC	China
PWID	Sadarghat	HIV-28	32	C	Bangladesh	CRF64_BC	China
PWID	Sadarghat	HIV-29	31	C	Bangladesh		
PWID	Sadarghat	HIV-30	26	CRF08_BC	China		
PWID	Sadarghat	HIV-31	63	C	China	C	China
PWID	Sadarghat	HIV-34	30	C	Bangladesh	CRF08_BC	Nepal

^a SBSW (Street-based Sex Worker), BBSW (Brothel-based Sex Worker), HBSW (Hotel-based Sex Worker), PWID (People Who Inject Drugs), PWID_F (Female PWID).

^b Drop-in center.

^c Laboratory identification number.

^d Age of participant.

^e Genotype based on NCBI HIV genotyping tool (2009 Ref Set, Window size 200, and step 50).

^f Country of the most similar strain by BLAST search (search done in October 2019).

7742) for primary strand synthesis and ED31F and ED633R (6798–7487) or 6559F and 7373R (6559–7373) for nested PCR. Again, if failed, we used alternative primers 5550F and 8445R (5550–8445) for primary strand synthesis and 5861F and ED12R (5861–7742) for nested PCR. With secondary PCR product ED5F and ED633R (6517–7487) or 6559F and ED633R (6559–7487) primers were used for 3rd round amplification.

Pol gene amplification

Viral RNA was extracted from plasma samples using the QIAamp Viral RNA Mini Kit according to the manufacturer's protocol (QIAGEN, Leusden, Netherlands). The entire protease gene

(99 amino acids) and at least the first 240 amino acids that encoded the reverse transcriptase (RT) gene were amplified by a nested RT-PCR using the One-Step RT-PCR kit from AgPath-ID (Applied Biosystems, California, USA) and QIAGEN HotStarTaq Master Mix kit with a specific primer (Table 1).

Sequencing and sequence analysis

All PCR products were analyzed by using agarose gel electrophoresis staining with GelRed™ (Biotium, Inc., CA) and purified using ExoSAP-IT kit (Life Technology, Foster City, CA). Sequencing was performed in an Applied Biosystems® 3500 Genetic Analyzer by using Big Dye® Terminator v3.1 Cycle

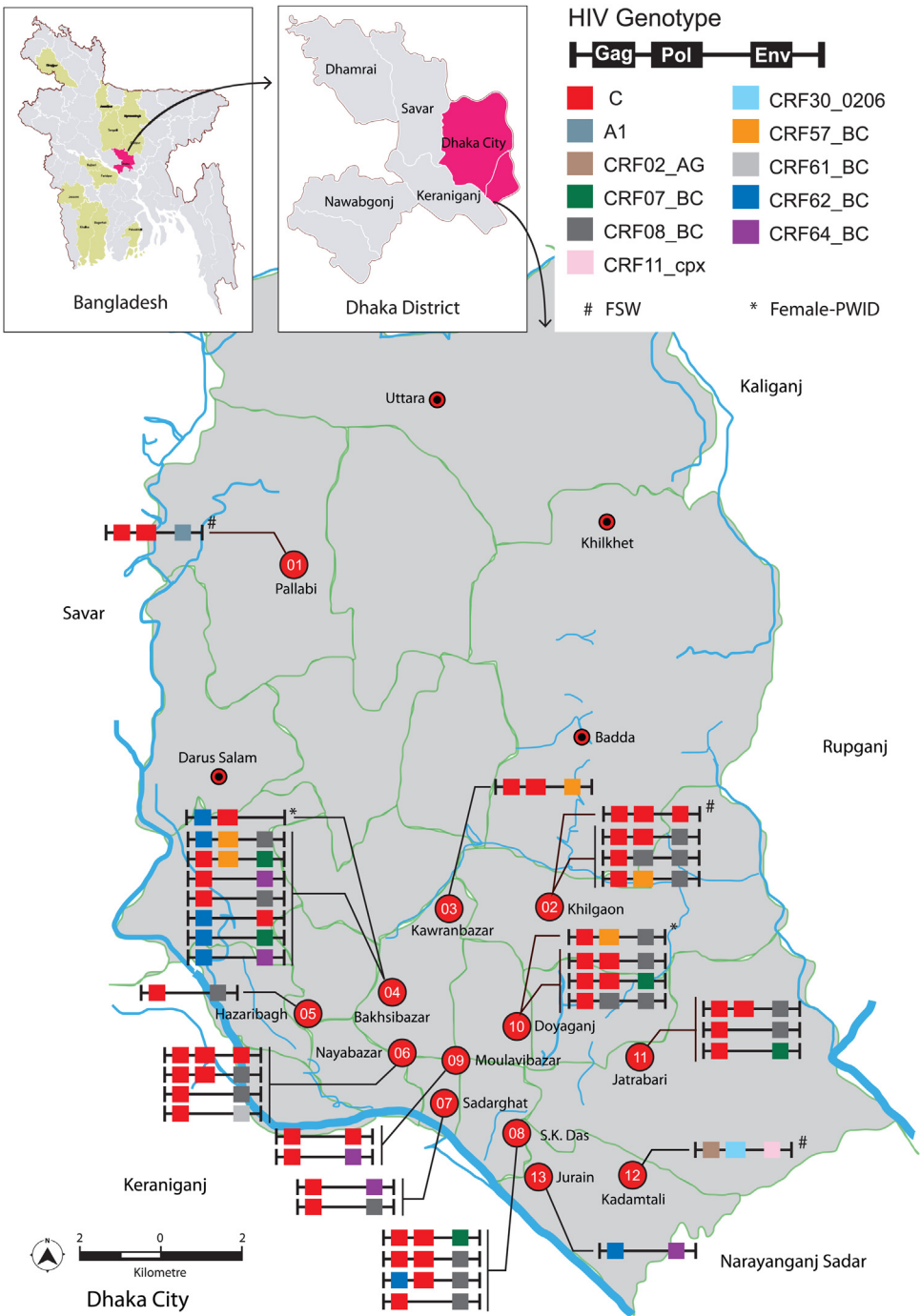


Figure 2. Genotype distribution of successfully sequenced sample from capital city Dhaka, Bangladesh.

Sequencing Ready Reaction kit (Life Technology). The chromatogram sequencing files were inspected using Chromas 2.23 (Technelysium, Queensland, Australia), and the consensus sequences were prepared using SeqMan II (DNASTAR, Madison, WI).

Sequence similarity search was performed using BLAST (blast.ncbi.nlm.nih.gov) to identify most similar global strains. HIV

subtyping was done using HIV genotyping tools (2009 Ref Set) in the National Centre for Biotechnology Information (NCBI; National Institutes of Health, Bethesda, MD) and p-distance of study sample with reference sequences. All sequences generated in this study were submitted to GenBank database under the accession Nos.: **MN720756–MN720911**.

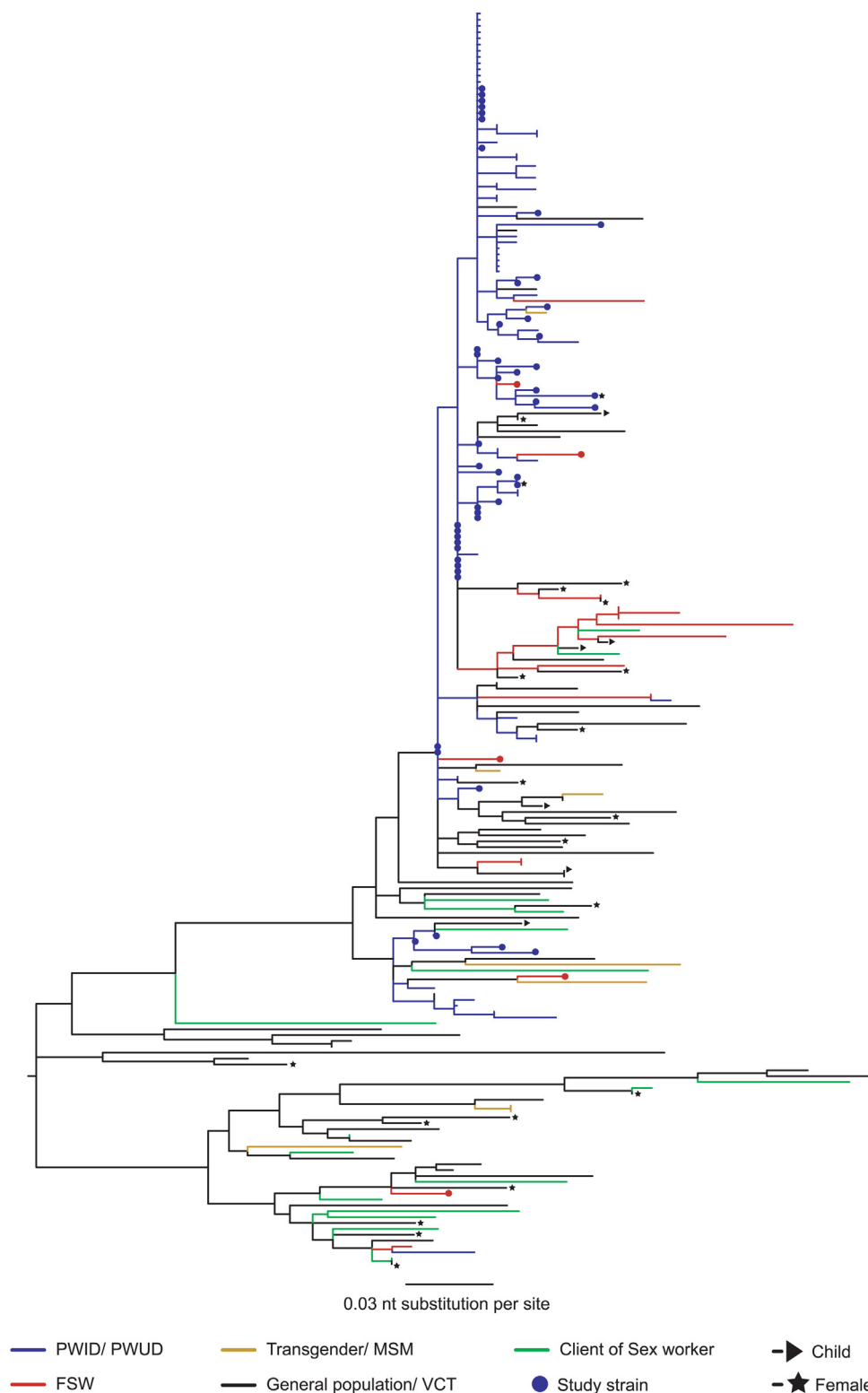


Figure 3. Phylogenetic analysis of Bangladeshi HIV-gag sequences. HKY + G + I substitution model was used with the 1000 bootstrap method in MEGA-X (version 10.0.5).

Phylogenetic analysis

A phylogenetic tree was constructed with total 241 Bangladeshi *gag* sequences (57 from the present study, 161 from GeneBank, and 23 sequences from other unpublished studies conducted by our laboratory). Sequences were aligned by built-in “ClustalW Multiple alignment” program of BioEdit (version 7.0.9.0) with 1000 bootstrap NJ method. The phylogenetic analysis was performed using MEGA-X (version 10.0.5). The Hasegawa-Kishino-Yano (HKY) model with evolutionarily Gamma distribution (G) parameter (HKY + G) substitution model was used with the 1000 bootstrap method; as this was found to be the most appropriate according to built-in “Find Best DNA/Protein Model” program (Kumar et al., 2018). By using study strains, a set of phylogenetic trees for *gag*, *pol*, and *env* was constructed to determine the genetic linkage between HIV-positive KPs from different Drop-in-Centers (DICs).

Drug resistance analysis

Genotypic drug resistance profiles were obtained from the Stanford University HIV drug resistance database (Program Version 2.2.11, Algorithm Version 8.8, <http://hivdb.stanford.edu/>). At the same time, the quality of *pol* gene was also assessed using the HIVseq programs, and HIValg program was used for the comparison of drug resistance profile with: i. Rega Institute (KU Leuven; v10.0.0), and ii. Agence Nationale de Recherches sur le SIDA (ANRS; v29).

Results

The study population consisted of HIV-positive individuals from the National HIV sero-surveillance, 2016, which included 60 male PWID, 7 female PWID, and 7 FSW. The male PWID were 26–63 years old, the female PWID were 25–56 years, and FSWs were 20–40 years old. Approximately 67% of male PWID and all female PWID were from the southern part of Dhaka. Among seven FSW, five were from Dhaka, one from Hili, and one from Tangail.

HIV genotype

A total of 65 (87.8%) samples were successfully sequenced either for *gag* or *env* gene. Among them, 64 *gag* and 56 *env* genes were successfully sequenced and genotyped, while 55 samples had both *gag* and *env* genes sequences (Table 2). The *gag* and *env* genes were amplified from proviral DNA, which was the basis for genotyping. Based on the *gag* gene, all HIV strains identified in our study were C (n = 47) except seven CRF62_BC and one CRF02_AG. On the other hand, based on *env* all except seven strains were recombinant forms of subtype B and C; CRF08_BC (n = 35), CRF07_BC (n = 5), CRF64_BC (n = 6), CRF61_BC (n = 1), and CRF57_BC (n = 1). Seven exceptions were C (n = 5), CRF11_cpx (n =

1), and A1 (n = 1). The genotype diversity of sequenced strains from Dhaka city is shown in Figure 2. All DICs were located in the southern part of the town, and a similar genetic pattern was identified in different DICs due to their geographical proximity.

Phylogeny

Phylogenetic analysis, which includes *gag* sequences of HIV strains collected over time from different KPs in Bangladesh, showed that most of the study PWID strains were placed into a large cluster and clearly separated from other risk group populations: FSW, transgender, and MSM (Figure 3). It was also observed that some PWID strains clustered sporadically with other previously identified PWID strains, though most of the strains generated in this study clustered in a monophyletic branch. This confirms that recurring transmission occurred within the PWID over time. On the other hand, strains from sex workers clustered with people who buy sex, travel outside of the country, and other KPs. For limited occasions, some strains from sex workers and PWIDs were closely related, which indicates spillover between these groups. A cluster of male and female PWID and the general population were also found, indicating limited sharing of strains between these populations.

A phylogenetic analysis of HIV strains obtained from different DICs showed distinct PWID clusters. Moreover, HIV strains from one cluster of the *gag* gene clustered differently for the *pol* or *env* gene (Figure 4). For example, the HIV-45 *gag* gene of DIC-08 (red in Figure 4) clustered with HIV-20, HIV-23, and HIV-34 of DIC-07, but the *pol* gene of the same strain (HIV-45) clustered with other *gag* (Figure 4A) and *env* (Figure 4B) genes such as HIV-50, HIV-62 of DIC-08, HIV-28 of DIC-07, HIV-37 of DIC-06, and HIV-09 of DIC-10 (Cyan in Figure 4).

Drug resistance and mutation in *pol* sequence

The *pol* genes were used for drug resistance analysis using viral RNA sequences (n = 10). All strains showed susceptibility against protease inhibitors (PIs) drug: atazanavir/ritonavir (ATV/r), darunavir/ritonavir (DRV/r), fosamprenavir/ritonavir (FPV/r), indinavir/ritonavir (IDV/r), lopinavir/ritonavir (LPV/r), and saquinavir/ritonavir (SQV/r) using three drug resistance algorithms (HIVdb, ANRS, and REGA) (Supplementary File 1). However, all showed resistance against tipranavir/ritonavir (TPV/r) by ANRS algorithm as they contained resistance-associated mutations (RAMs) M36IV, H69K, and L89M in the *Protease* region.

All RT sequences were susceptible to NRTIs (nucleoside/nucleotide reverse transcriptase inhibitors) drugs, which include: abacavir (ABC), zidovudine (AZT), stavudine (D4T), didanosine (DDI), emtricitabine (FTC), lamivudine (3TC), and tenofovir disoproxil fumarate (TDF) and also nonnucleoside reverse transcriptase inhibitors (NNRTIs) drugs; delavirdine (DOR or DLV),

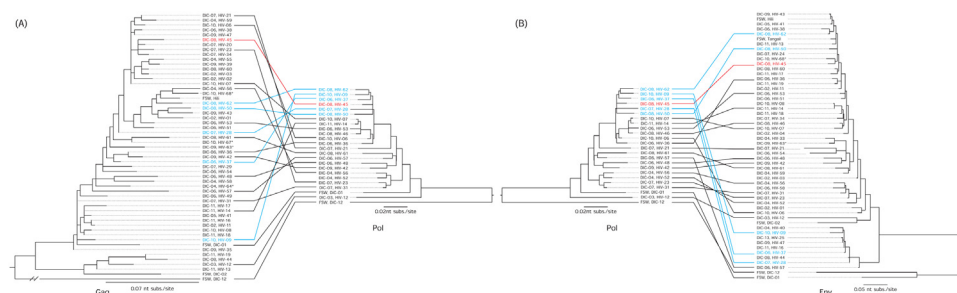


Figure 4. Phylogenetic tree of HIV strains obtained from different drop-in centers. (A) Comparative phylogenetic relationship between *gag* and *pol* sequences. (B) Comparative phylogenetic relationship between *pol* and *env* sequences.

efavirenz (EFV), etravirine (ETR), nevirapine (NVP), and rilpivirine (RPV).

Discussion

This study represents *gag* and *env* sequences that allowed us to identify HIV transmission dynamics among the KPs of Bangladesh. We also report here *pol* sequence-based drug resistance profile of HIV-1 strains circulating in Bangladesh. To our knowledge, this is the first HIV drug resistance report from Bangladesh. Here, we provide evidence that: (i) HIV subtype C-related recombinants have remained the major circulating genotypes in Bangladesh, (ii) recurring transmission of subtype C-related recombinants occurred within PWID, (iii) limited spillover from PWID to sex workers has been occurred, (iv) Bangladeshi HIV strains are susceptible to NRTIs and NNRTIs drugs, and (v) a unique recombination form (URF) with genotype C for *gag-pol* and A1 for *env* have been identified.

Inter-subtype recombinant genomes known as the unique recombinant form (URF) are common in HIV, generated through mixed infections. If a URF is identified among too many people, it becomes one of the circulating strains in the HIV epidemic and is recognized as the “circulating recombinant form (CRF).” All HIV strains identified in our study were CRF except one URF, which indicates that most of the CRF strains were already in circulation in different parts of the world, including Bangladesh. These CRFs were recombinants between subtypes C and B.

Only two strains were non-subtype C related recombinants: AG recombination (HIV-74) and a URF between subtype C and A1 (HIV-70). The AG recombinant was identified earlier in STI client and returnee migrant workers (Sarker et al., 2008; Ljungberg et al., 2002) and clustered with other A related recombinant strains obtained from voluntary counselling and testing (VCT) unit clients (Urmi et al., 2015), where one of them was a client of a sex worker (Figure 3). The URF recombinant was a mixture of subtype C for *gag-pol* and subtype A1 for *env* region, which was supported by our phylogenetic analysis and was similar to another URF described recently in China among HIV/HBV/ HCV triple-infected PWID (Deng et al., 2019). Mobility behavior from the BSS data showed that 14.6% of PWID traveled to different districts while few traveled to India and bought sex while abroad (NASP, 2017). Nepal reported high HIV prevalence among returnees from India, particularly FSW (44%) (Zhang et al., 2017). In contrast, China reported a low HIV prevalence (0.2%–0.6%) among cross border travelers from Myanmar and Vietnam (Wang et al., 2017; Zhang et al., 2017). Bangladesh has an extensive trading relationship with its neighboring countries, and therefore, cross-border HIV transmission appears to be very common in this region (Table 2).

Most of the *gag* sequences from PWID in this study were clustered with the same population group, though few of them clustered sporadically with the strain obtained from other KPs (Figure 3). Also, PWID strain clustered differently for individual gene segments (Figure 4). This type of phylogenetic cluster of PWID from different DIC analyses provides evidence of genetic mix-up among PWID from different DICs. On the other hand, DICs of old Dhaka (the epicenter of HIV) are geographically very close to each other (Figure 2). Therefore, insubstantial address, movement behavior, and unchanged needle/syringe sharing practices of PWID may cause the HIV epidemic among PWID of Dhaka city (NASP, 2017).

Behavioral data showed that 31% of male PWID bought sex from FSW, of whom 45% did not use condoms (NASP, 2017), and most of the female PWID were involved in transactional sex (Azim et al., 2015). Therefore, a spillover between FSW and PWID likely occurred (2/5 FSW were placed in PWID cluster) (Figure 3). A spillover between FSW and PWID increased the risk of HIV

transmission in the general population (de Oliveira et al., 2017; Dyer et al., 2017; Li et al., 2018; Paraskevis et al., 2018). A similar finding was observed in this study with a cluster of the general population; PWID, sex worker, and clients of the sex worker may result from this spillover and occur through the sexual transmission of HIV. But the social network, particularly the sexual and injecting network of PWID, has never been systematically studied in Bangladesh. Also, HIV-positive individuals who did not disclose their behavioral status due to fear of social stigma was classified as the general population, which was a barrier to define the transmission network between KPs. As a result, current HIV interventions may not reach the critical component (i.e., “core-transmitters”), who can be the source of the continuous spread of HIV.

The government of Bangladesh provides ARV drugs to the PLHIV through AIDS/STD program (ASP) in collaboration with other NGOs since 2012. Presently, different types of drugs or drug combinations are available in Bangladesh (Ahassan et al., 2018), and according to National AIDS and STD Program, ~4500 PLHIV are on ART (AIDSdatahub, 2019). Although the ARV resistance test is essential to initiate ART and select appropriate regimens accordingly (DHHS, 2019), ART initiation does not depend on the CD4 count or viral load; “test and treat strategy” is followed to initiate ART.

This is the first report of ART drug-resistant profile from Bangladesh with limited sample size and should therefore be considered with caution. However, this study found that HIV strains in Bangladesh are susceptible to NRTIs and NNRTIs available in Bangladesh. Interestingly, the ANRS algorithm showed that all tested samples were resistant to tipranavir/ritonavir (TPV/r) drugs that are unavailable in Bangladesh. TPV/r, a protease inhibitor, was evaluated as an initial therapy for treatment-naïve HIV-1-infected patients because of its potency, unique resistance profile, and high genetic barrier (Gathe et al., 2006). Therefore, the careful use of tipranavir/ritonavir (TPV/r) drugs for HIV treatment in Bangladesh is suggested.

This article has some limitations. First, we did not sequence the full genome of HIV strains and provided genotype data based on the partial *gag* and *env* sequences, which may fail to illustrate true genotype data to identify CRF. Second, we could not amplify 13% *gag* and 24% *env* segments of proviral DNA from whole blood samples and 82% *pol* segment of viral RNA from the serum sample. The reason underlying this could be manifold. According to surveillance criteria, the positive samples were diagnosed based on antibody status and were unlinked and anonymous. Therefore, we had no idea about the viral load, CD4 count, and ART history, which could impact the amount of viral RNA, e.g., individuals with high CD4 count might have a low viral load. Finally, our data may fail to provide a true scenario of drug resistance data and genotypic information of Bangladeshi circulating strains due to the small sample size. Despite these limitations, the study findings warrant continuous monitoring of HIV-positive individuals and future investigation to identify social networks within and between KPs to halt the transmission and prevent new infections.

Authors' contributions

MusR and MSS conceived and coordinated the study and wrote the paper. SR performed laboratory procedures, including diagnostic testing, sequencing and genotyping, data analysis, interpretation of the results, and drafted the first manuscript. SIK involved in critically reviewing the manuscript and provided intellectual input. MusR, MSS, and MasR reviewed the article critically for intellectual content. All authors reviewed subsequent drafts of the manuscript and approved the final version.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.ijid.2020.12.037>.

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